Making Sense of Probabilistic Inference: A Weeks 1-5 Rebuild · Quiz explanations

Original W6 formative checkpoints; probabilistic identities follow the reading map in the lesson.

1. Does drawing 1,000 more posterior samples give us 1,000 new sensor observations?

No. The observations stay fixed. Samples are simulated possible hidden states used to calculate an answer.

2. What does the normalizing denominator add up?

The joint weights of every possible hidden path with the observed readings held fixed. Their total is the evidence, 0.0872 in this example.

3. Why is the probability of the MAP path different from the probability that the last state is one?

The MAP path is one complete event, 111. The last-state event includes 001, 011, 101 and 111; their probabilities must be added.

4. At time 2, why must an online particle filter be compared with a filtering reference rather than the full-data smoother?

They must use the same information. The filter uses readings 1 and 2, while the smoother also uses reading 3.

5. Can more optimization always remove the error caused by a factorized variational family?

No. Optimization changes parameters within the family. It cannot produce dependence that the family excludes; local optimization can introduce an additional gap.

6. Which uncertainty shrinks when we draw more samples while keeping the observed data fixed?

Typical numerical error of the estimator shrinks. Posterior spread remains fixed. The usual inverse-square-root sample-count formula assumes independent draws and finite variance.

7. Why must a rejected MH proposal remain as a repeated state in the chain?

The chain stays in its current state on rejection. Its residence times contribute to the intended stationary distribution; deleting repetitions changes the represented distribution.

8. What must stay fixed to interpret a comparison as an algorithm comparison?

The observed data, model parameters, target and query. Also record budgets, seeds and the metric; changing the target answers a different comparison question.